Clarisa Caballero-Ignacio

M.Eng. Computer Science Student | Research | Program Administrator

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Education

Master of Engineering, Computer Science
Oregon State University — Focus: Cybersecurity; Minor: Artificial Intelligence

Bachelor of Science, BioResource Research Interdisciplinary Science Program
Oregon State University — Focus: Toxicology; Minor: Spanish

Professional Experience

Coordinator of Student Outreach and Scholarships
OSU College of Agricultural Sciences | Academic Programs Office

Coordinator of Student Outreach and Retention
OSU College of Agricultural Sciences | Academic Programs Office

Faculty Research Assistant
OSU Food Safety and Environmental Stewardship Program | Analytical Chemistry

Skills

Programming: Python, C, C++
Web Development: HTML, CSS, JavaScript, Drupal
Data & Analytics: Power BI, Excel
Research & Laboratory: GC-MS, LIMS, Liquid Extraction, QA/QC, Quantitative Analysis
Technical Tools: GitHub, LaTeX
Administrative Systems: Microsoft Office Suite, Banner, Core Reports, Beaver Hub, Scholarship Universe, Blackbaud
Languages: English (fluent), Spanish (fluent)

Research Publications

Caballero-Ignacio, C., Stephens, L., Sage, J., Motwani, M., & Ma, Z. "Vulnerability Trends Across Open-Source AI-Attributed Code." *To appear in the Proceedings of the 2nd Workshop on Explainable and Reliable Software Systems (EXPRESS '26), 2026.*

Research Experience

Graduate Research
Oregon State University | PI: Zane Ma, Ph.D.

- First author of *Vulnerability Trends Across Open-Source AI-Attributed Code*, a peer-reviewed empirical study published in the proceedings of the 2nd Workshop on Explainable and Reliable Software Systems (EXPRESS '26).
- Analyzed 253,390 AI-attributed and 9,544 human-written source code using CodeQL and statistical analyses. Identified significant differences in security-relevant CWE prevalence across AI tools, programming languages, and human-written Python.

Undergraduate Thesis
Oregon State University | PI: Kathy Magnusson, Ph.D.

- Investigated the role of the gut microbiome in diet-related cognitive changes and performed behavioral assessments evaluating memory, anxiety, impulsivity, and cognitive flexibility using the Morris water maze, marble burying, step-down latency, and novel object behavioral tests.
- Findings: An antibiotic cocktail improved long-term memory independent of diet.

- Thesis: *Assessment of Gut Microbiome Role in Diet-Related Changes in Cognition.* ScholarsArchive@OSU

Grants

- **USDA NIFA Multicultural Scholars Program** ($250,000) 2023
  Co-Principal Investigator, College of Agricultural Sciences, BioResource Research
- **USDA NIFA Multicultural Scholars Program** ($200,000) 2020
  Grant Project Lead, College Assistance Migrant Program (CAMP)

Teaching

Co-Instructor - AGRI 399/599: Discovering Ancient and Modern-day Agriculture in the Heart of Mexico

Academic Programs Office, Oregon State University

- Co-instructed a short-term study abroad program in Mexico for 12 undergraduate and graduate students. Partnered with the Oregon State University SMILE Program to recruit students, secure funding for 10 participants who developed K-12 agriculture-focused educational materials, and coordinated program logistics.

Student Outreach, Mentoring, & Inclusive Excellence

MANRRS & SACNAS Chapter Advisor

College of Agricultural Sciences, Oregon State University

- Increased chapter membership by more than 170% through analysis of engagement metrics, targeted outreach strategies, and continuous assessment of program effectiveness.
- Develop and oversee mentorship programming, conference participation opportunities, and leadership development activities.
- Built a data-driven tracking and reporting system to monitor recruitment, retention, and student engagement trends.

USDA NIFA Multicultural Scholars Program Mentorship Lead

College Assistance Migrant Program (CAMP), College of Agricultural Sciences

- Led implementation of mentorship programming and student engagement initiatives in collaboration with the Principal Investigator.
- Supported student development through mentoring activities, program coordination, and retention efforts.

Awards & Honors

- **State of Inclusive Excellence** 2026
  Oregon State University
- **Club Advisor of the Year** 2025
  College of Agricultural Sciences
- **SACNAS Chapter of the Year Award** 2025
  Society for Advancement of Chicanos/Hispanics and Native Americans in Science
- **MANRRS Chapter of the Year Award** 2025
  Minorities in Agriculture, Natural Resources, and Related Sciences
- **Diversity Commitment Award — MANRRS Chapter at OSU** 2025
  College of Agricultural Sciences
- **Diversity Commitment Award — SACNAS Chapter at OSU** 2023
  College of Agricultural Sciences
- **FSES Challenge Coin** 2018 & 2019
  Food Safety and Environmental Stewardship Program

References available on request.